TED(21)-5072 **REVISION 2021**

MODEL QUESTION PAPER

PROCESS CONTROL AND INSTRUMENTATION

Max.Marks : **75**

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. List any two process variables in chemical plants.
2. State dead zone.
3. Identify any two types of thermocouples used to measure temperature.
4. State the principle of operation of the Coriolis flow meter.
5. Outline the function of an I to P converter
6. SCADA is the abbreviation for
7. List one example for two position control.
8. Define chromatography.
9. Identify an instrument to study the surface morphology of a material.

Part B

II. Answer any EIGHT questions from the following. Each questions carries 3 marks

8 X 3 = 24 Marks

1. Elucidate the role of process control in chemical plants.
2. Identify the components of a control system.
3. Illustrate the working of a bourdon tube pressure gauge.
4. Summarize the working of radar tank gauging system.
5. Discuss how a conductivity meter measures conductivity.
6. Illustrate proportional control.
7. Summarize Distributed Control System (DCS)
8. Illustrate the working of solenoid valves
9. Outline the classification of spectrometers
10. Summarize the working of a transmission electron microscope.

Part C

Answer ALL questions. Each question carries 7 marks

6 X 7 = 42 Marks

III. Illustrate a simple level control system

OR

IV. Summarize the elements of a measuring instrument.

V. A furnace in a chemical industry is operating above 4500 ^{o}C. Suggest a temperature measuring instrument to measure this temperature and elucidate its working.

OR

VI. Outline the working of an ultra-sonic flow meter.

VII. Demonstrate how a wet and dry bulb thermometer is used to determine humidity.

OR

VIII. Summarize the working of differential pressure transducers used in venturi and orifice flow meters.

IX. Differentiate servo and regulatory control

OR

X. Outline Programmable Logic Controllers [PLC]

XI. Differentiate the valve safe positions Fail Open (FO) and Fail Close (FC) with an example.

OR

XII. Discuss cascade control system in process industries

XIII. Illustrate Liquid Solid Chromatography.

OR

XIV. Suggest any one suitable method for the size determination of nano particles and elucidate its principle of operation.